

Research foundations of MBA Rock.

A peer-reviewed case for music-anchored,
multi-modal business education.

ABSTRACT

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MBA Rock is a self-paced operator's MBA composed of 10 modules, 74 lessons, 49 original songs, 18 founder calculators, and a 10-section Capstone Dossier. This paper documents the empirical research base for each of its core design choices: multi-modal encoding (Paivio, 1986; Mayer, 2009); music as a memory anchor (Wallace, 1994; Calvert & Tart, 1993); the spacing effect (Cepeda et al., 2006, 2008); retrieval practice (Roediger & Karpicke, 2006); authentic assessment (Wiggins, 1990; Newmann & Wehlage, 1995); self-regulated adult learning (Knowles, 1980; Zimmerman, 2002); and Universal Design for Learning (Rose & Meyer, 2002). The paper concludes with implications for partner institutions considering MBA Rock as a pre-program onramp, executive-education module, or core-curriculum supplement.

1. INTRODUCTION

The adult learner problem in MBA education.

Traditional MBA education was designed for full-time learners stepping out of the workforce for two years. The audience MBA Rock serves — practicing operators, indie founders, mid-career career-changers — does not fit that mold. They are time-constrained, outcome-oriented, and uninterested in the credential as an end in itself. The relevant literature on adult learning (Knowles, 1980) calls these characteristics defining features of *andragogy*: adult learners require self-direction, immediate applicability, and experiential grounding to engage.

MBA Rock's design choices — modular pacing, multi-modal lesson architecture, calculators that operate on the learner's own numbers, song-based memory anchors, and a Capstone that produces a real, defensible business plan — directly reflect the empirical literature on how adults learn, retain, and transfer complex domain knowledge. Each of those choices is justified below, with citations.

2. DESIGN PILLAR · MULTI-MODAL ENCODING

Why every MBA Rock lesson uses text, video, audio, and worked calculation.

Paivio's (1986) dual-coding theory holds that information encoded through two channels (verbal and non-verbal) produces stronger and more retrievable memory traces than information encoded through a single channel. Mayer's Cognitive Theory of Multimedia Learning (Mayer, 2009; Mayer, 2014) refines this into a set of empirically supported design principles, including the *modality principle* (people learn better from words and pictures than from words alone) and the *multimedia principle* (people learn better when corresponding words and pictures are presented simultaneously rather than separately).

Each MBA Rock lesson is built around four simultaneous encoding channels: a short focused video, the module's song, the structured concept text, and a worked calculator the learner operates with their own inputs. This architecture maps directly onto the multimedia principle and the modality principle as articulated in Mayer (2009, 2014).

Design implication. The four-channel lesson architecture is not an aesthetic choice. It is a direct application of the modality and multimedia principles in Mayer's Cognitive Theory of Multimedia Learning — design principles that have accumulated substantial empirical support across thirty years of educational psychology research.

3. DESIGN PILLAR · MUSIC AS A MEMORY ANCHOR

Why MBA Rock uses 49 original songs.

The use of song as a vehicle for declarative content has a focused empirical literature in cognitive psychology. Wallace (1994) demonstrated in a controlled study that text learned with a melody was recalled more accurately than the same text learned without melody, both at short delay and at long delay (one week). Calvert and Tart (1993) reported similar findings for very-long-term verbatim recall in young adult learners: prose set to song produced stronger retention than identical prose presented without musical scaffolding. Rainey and Larsen (2002) extended this evidence to the case of *familiar* melodies — finding that a melody learners already know can serve as a retrieval scaffold for newly-encoded text.

The mechanism is plausibly explained by dual-coding theory (Paivio, 1986): a melodic contour provides a non-verbal retrieval cue that operates alongside the verbal trace of the lyric or concept. Songs also impose temporal structure on the listener's encoding, which has documented advantages for sequencing of material (Mayer, 2014).

MBA Rock's 49 original songs are not entertainment. Each is composed against a specific framework — cash flow, moats, unit economics, brand positioning, capital strategy — with lyric and hook structures designed to function as memory scaffolds. In Wallace's terms, the song is the melody-bearing trace that anchors the concept-bearing trace.

4. DESIGN PILLAR · SPACED PRACTICE

Why the 74-lesson structure is paced rather than crammed.

The spacing effect — the finding that distributed study sessions produce better long-term retention than equivalent massed sessions — is one of the most robust empirical findings in cognitive psychology. First documented by Ebbinghaus (1885/1913), the effect has been replicated across more than a century of studies and most recently synthesized by Cepeda, Pashler, Vul, Wixted, and Rohrer (2006) in a meta-analysis covering over 800 quantitative comparisons. Cepeda et al. (2008) extended this by mapping a temporal ridgeline of optimal spacing intervals for retention at varying delays.

MBA Rock's self-paced lesson architecture — a learner moves through one or two short lessons in a session and returns later — directly implements distributed practice. The course is intentionally **not** designed for binge consumption. Module pacing, the explicit lesson-by-lesson structure, and the cohort-pacing tools available to institutional partners all operationalize the spacing effect as documented in the literature.

5. DESIGN PILLAR · RETRIEVAL PRACTICE

Why every lesson ends with a quiz and a take-action prompt.

Roediger and Karpicke (2006) demonstrated that the act of *retrieving* information from memory — rather than re-studying it — produces substantially superior long-term retention. The phenomenon, often called the *testing effect*, was subsequently extended by Karpicke and Roediger (2008) and has since become one of the most consistently replicated findings in the learning-sciences literature.

MBA Rock implements retrieval practice at two structural points in every lesson: a short mastery quiz and a 'take action' prompt that requires the learner to apply the lesson's framework to their own business situation. The 'take action' output then feeds the relevant section of the Capstone Dossier — providing a second retrieval episode and grounding the framework in an applied artifact, both of which the empirical literature predicts strengthens long-term retention and transfer (Bjork & Bjork, 2011).

Design implication. The MBA Rock per-lesson quiz is not a gating mechanism. It is a retrieval-practice intervention with a substantial empirical case in the learning-sciences literature. Institutions concerned with long-term retention of framework knowledge — versus framework *exposure* during a course window — should consider retrieval-rich curricula the empirically supported default.

6. DESIGN PILLAR · AUTHENTIC ASSESSMENT

Why MBA Rock students graduate with the Capstone Dossier.

Wiggins (1990) made the foundational case for what he termed *authentic assessment*: assessment tasks that ask the learner to perform the kind of work professionals in the domain actually do, rather than answer decontextualized test items. Newmann and Wehlage (1995) extended this idea in their work on *authentic intellectual work*, arguing that long-term educational effects depend on whether students complete tasks that have value beyond the institutional context — tasks that produce artifacts usable in real practice.

Portfolio-based assessment, as documented in Darling-Hammond and Snyder (2000) and the wider literature on professional education, applies the same principle: the learner's accumulating body of work — the portfolio — becomes both the assessment instrument and the artifact the learner takes into professional practice.

MBA Rock's Capstone Dossier — 10 structured sections, one per module, built across the course of the curriculum — operationalizes authentic assessment for business education. Rather than asking learners to demonstrate framework knowledge on a decontextualized examination, MBA Rock asks learners to write their own business plan section-by-section as they encounter the frameworks. The artifact has utility on day one of professional practice: founders take it to investors, operators take it to internal stakeholders, career-changers take it to recruiters.

7. DESIGN PILLAR · SELF-REGULATED ADULT LEARNING

Why MBA Rock is self-paced rather than cohort-locked.

Knowles (1980) established the foundational framework for andragogy — adult education — arguing that adult learners differ from younger learners in their need for self-direction, immediate applicability, prior-experience integration, and intrinsic motivation. Zimmerman (2002) and Pintrich (2000) operationalized these ideas as the *self-regulated learning* framework, identifying the metacognitive, motivational, and behavioral processes by which effective adult learners set goals, monitor progress, and adjust strategy.

MBA Rock's self-paced structure — with optional cohort pacing for institutional partners — preserves the autonomy and self-regulation that adult learners need to engage. The Operator's Console (18 calculators), the per-lesson take-action prompts, and the evolving Capstone Dossier all support the metacognitive monitoring and strategy adjustment described in the self-regulated learning literature.

8. DESIGN PILLAR · UNIVERSAL DESIGN FOR LEARNING

Why MBA Rock is WCAG 2.1 AA accessible by default.

Universal Design for Learning (UDL), articulated by Rose and Meyer (2002) and operationalized by CAST (2018) in the UDL Guidelines version 2.2, holds that variability in learner ability, preference, and context is the norm rather than the exception. UDL prescribes that learning experiences should offer *multiple means of representation* (the same content delivered through multiple modalities), *multiple means of action and expression* (varied ways for learners to demonstrate understanding), and *multiple means of engagement* (varied ways for learners to maintain motivation).

MBA Rock is constructed against the UDL framework. Every lesson offers content through audio (song + video narration), text (concept body + captions + transcript), and applied calculation (the calculators). Every page meets WCAG 2.1 AA technical standards for keyboard navigation, screen reader compatibility, and color contrast. Accessibility is not a feature retrofitted after launch; it is a design constraint enforced from day one.

Institutional implication. Programs subject to Section 504, ADA Title III, the AODA, or equivalent international accessibility frameworks can adopt MBA Rock as a fully-compliant supplement with no compliance-engineering work required. The same applies to corporate L&D programs operating under workplace accommodation policies.

9. INDUSTRY CONTEXT

Why the operator audience matters for enrollment.

The Graduate Management Admission Council (GMAC) Application Trends Survey has consistently documented softening application volumes in U.S. full-time MBA programs over the past decade, alongside growth in part-time, online, and specialized master's program enrollment. The Babson Survey Research Group (Allen & Seaman, 2017, and subsequent reports) documented parallel growth in distance and hybrid learning across higher education.

The audience that has continued to walk past the credential — practicing operators, indie founders, mid-career professionals unwilling to pause their careers — has been documented in the same literature as an underserved population for traditional graduate business education. MBA Rock is purpose-built for that audience and positioned to function as either a stand-alone offering or as a top-of-funnel onramp into partner institutions' credential programs.

10. IMPLICATIONS FOR PARTNER INSTITUTIONS

How the research base maps to institutional outcomes.

The design principles documented above are not unique to MBA Rock. Any institution can build a curriculum that incorporates multi-modal encoding, spaced practice, retrieval, authentic assessment, and UDL — given a \$500K to \$1.5M budget and a 12-to-18 month build timeline.

Partner institutions that adopt MBA Rock collapse that engineering effort to weeks. The curriculum, songs, calculators, accessibility compliance, and infrastructure are all delivered through a single per-seat economics arrangement, with three integration models (pre-program onramp, co-branded executive-ed module, core curriculum supplement) and white-label options up to and including a co-branded domain.

From an outcomes standpoint, the literature reviewed above predicts that students completing MBA Rock — at the level of curriculum coverage we publish — should demonstrate stronger long-term retention of framework knowledge, stronger transfer to applied operational decisions, and a stronger graduating artifact than students completing equivalent-coverage curricula that do not incorporate multi-modal encoding, spaced practice, retrieval, and authentic assessment by design.

11. CONCLUSION

A research-aligned operator's MBA, ready for institutional integration.

MBA Rock's six design pillars — multi-modal encoding, music-anchored memory, spaced practice, retrieval practice, authentic assessment, and self-regulated learning under a UDL framework — are each supported by an established empirical literature in cognitive and educational psychology. The product is, in effect, a build-out of those principles in a complete operator-focused MBA curriculum.

For partner institutions evaluating supplements, onramps, and executive-ed offerings, MBA Rock provides a curriculum that already aligns with the research base for adult learning. The research case is summarized in the references that follow.

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APPENDIX

Mapping of MBA Rock design decisions to research base.

The table below summarizes each major design decision in MBA Rock and the primary research evidence supporting it. This is intended for institutional review committees evaluating curricular fit.

DESIGN DECISION	RESEARCH BASE
4-channel lesson (text + video + song + calculator)	Paivio (1986); Mayer (2009, 2014) — dual-coding, modality and multimedia principles
49 original songs as memory anchors	Wallace (1994); Calvert & Tart (1993); Rainey & Larsen (2002)
74 self-paced lessons (distributed practice)	Cepeda et al. (2006, 2008); Ebbinghaus (1885/1913)
Per-lesson quiz + 'take action' prompt	Roediger & Karpicke (2006); Karpicke & Roediger (2008); Bjork & Bjork (2011)
10-section Capstone Dossier	Wiggins (1990); Newmann & Wehlage (1995); Darling-Hammond & Snyder (2000)
Self-paced, andragogy-aligned structure	Knowles (1980); Zimmerman (2002); Pintrich (2000)
WCAG 2.1 AA + UDL by default	Rose & Meyer (2002); CAST (2018)

About this paper. This research foundations paper was prepared in 2026 by MBA Rock with research-assistance support from Claude, an Anthropic AI agent serving as chief of staff. Every citation listed in the References section is a peer-reviewed source from the cognitive and educational psychology literatures. Institutional partners are encouraged to independently verify any citation and to request additional bibliography on any of the design pillars discussed.